

Jonathan Cruz

SENIOR FORWARD DEPLOYED ENGINEER · AI AGENTS & AUTOMATION

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PROFILE

"Senior engineer with 10+ years shipping consumer and B2B fintech products end-to-end. Daily practitioner of agentic AI engineering – Claude Code with sub-agent orchestration, custom skills, hooks, and project-level specifications – applied to real production codebases. Strong TypeScript across the stack (NestJS, React/React Native, Postgres) with deep experience in fintech compliance, payments, and event-driven systems. Comfortable embedding with non-technical stakeholders to translate manual processes into reliable automated workflows, and partnering across org lines to ship outcomes rather than tickets."

PROFESSIONAL EXPERIENCE

SELECTED ROLES & CONTRIBUTIONS

Sydecar

FULL-STACK SOFTWARE ENGINEER

Agentic AI Engineering

CLAUDE CODE IN PRODUCTION

Daily user of Claude Code with sub-agent orchestration for parallel exploration and review, custom skills for repeatable workflows, pre/post-tool hooks for guardrails, and project-level specifications encoding architectural rules. Apply spec-driven development for code generation, regression-style review, and test scaffolding — pairing LLM execution with knowledge-centric project context to keep iteration speed high without sacrificing quality.

CLAUDE CODE AGENTIC WORKFLOWS SUB-AGENTS SPEC-DRIVEN

Cross-functional Discovery

TRANSLATING MANUAL PROCESSES

Partnered directly with compliance officers to translate manual KYC/EDD workflows into reliable automated systems: LP funding blocks, UBO verification, FinCEN integration, CTA compliance, automatic KYC revalidation. Led discovery sessions, produced technical scopes (ADRs), and embedded with non-technical SMEs from requirements through production deployment.

COMPLIANCE KYC ADRS DISCOVERY

AI-driven Reliability

EVAL & GUARDRAILS PATTERNS

Built race-resistant verify-identity flow with re-check inside to handle "email-free-at-blur → taken-at-submit" edge case (409 EMAIL_NOW_EXISTS). Standardized webhook idempotency via (unique) with PROCESSING/PROCESSED/REJECTED state machine. Apply the same discipline — invariants, retries, audit-friendly state machines — to AI-driven flows.

IDEMPOTENCY CONCURRENCY AUDIT LOGGING

Event-driven Systems

OUTBOX + BULLMQ

Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base in ledger; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR with deterministic predecessor lookup. Patterns directly applicable to agentic pipelines requiring resilience and auditability.

BULLMQ OUTBOX REDIS EVENT-DRIVEN

External Integrations

SAAS APIS

Built bidirectional QuickBooks invoice sync engine with state validation, auto-sync on approval, and full E2E suite (Playwright, Cucumber BDD). Worked on Plaid Link integration with processor tokens powering ACH via Treasury Prime, including a mock banking engine for local sandbox. Comfortable owning end-to-end integration with third-party SaaS stacks.

PLAID TREASURY PRIME QUICKBOOKS REST APIS

Type-safe Architecture

NESTJS MONOREPO

Designed (ADR) and implemented Billing V2 with quarterly fee scheduler and billing model versioning. Migrated legacy controllers (auth, onboarding, admin) to type-safe ts-rest contracts with Zod schemas. Operate inside an Nx monorepo with dozens of NestJS libraries — same stack profile as modern microservice-oriented orgs.

NESTJS NX TS-REST ZOD

B2B EMBEDDED COMMERCE

Zax

SR SOFTWARE ENGINEER

JAN 2019 — MAY 2023
SÃO PAULO

Embedded Financial Workflows

MOBILE + WEB

Built from scratch a wholesale ordering app with integrated payment flows and credit-limit validation for business partners, delivering embedded financial services to thousands of frontline B2B users with React Native, TypeScript, and GraphQL.

REACT NATIVE

TYPESCRIPT

GRAPHQL

Cloud Infrastructure

AWS

Deployed full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) with serverless React/Next.js apps. Containerized services with Docker for consistent dev-to-prod parity.

AWS

NEXT.JS

DOCKER

SERVERLESS

Delivery Excellence

CI/CD

Configured end-to-end pipeline with GitHub Actions, Fastlane, and Code Climate for automated quality gates. Reduced release cycle from weeks to days.

GITHUB ACTIONS

FASTLANE

DOCKER

CONSUMER MOBILE FINTECH

Netlolo

SR SOFTWARE ENGINEER

SEP 2018 — JAN 2019
SÃO PAULO

Consumer Mobile Product

REACT NATIVE

Built a mobile and web product bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum. Shipped to non-technical end users with attention to onboarding and trust.

REACT NATIVE

TYPESCRIPT

WEB3

Real-time Systems

SDK

Developed a chat SDK for businesses using React Native and React/Redux, powered by XMPP and a Java backend. Designed for embedding into other teams' apps — same force-multiplier mindset as forward-deployed work.

REACT

REDUX

XMPP

JAVA

Zero

DOWNTIME

Built CI/CD automations for continuous deployment without service interruption. PostgreSQL with Hasura as the GraphQL layer.

POSTGRESQL

HASURA

CI/CD

CONSUMER HEALTHCARE

Escale

SR SOFTWARE ENGINEER

JAN 2018 — SEP 2018
SÃO PAULO

Empowering Other Engineers

TECH TALKS

Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture, coaching engineers across product teams. Established service-boundary and API-contract patterns adopted across the org — acting as a technical force multiplier inside a non-engineering-led environment.

MENTORSHIP

TECH TALKS

MICROSERVICES

Microservices

MULTI-DB

Built the product foundation with React, Redux, and a Node.js/TypeScript backend. Selected appropriate persistence per service responsibility (MySQL, PostgreSQL, MongoDB) and drove DB design decisions across teams.

REACT

NODE.JS

MONGODB

CIRCLECI

CONSUMER LENDING FINTECH

PraValer

PL SOFTWARE ENGINEER

JUN 2017 — JAN 2018
SÃO PAULO

Process Automation

LEGACY MODERNIZATION

Migrated legacy PHP loan origination platform to a Node.js/Express backend for a student loans product with governmental funding. Converted manual, error-prone steps into automated, auditable flows for loan request lifecycle, eligibility, and disbursement.

NODE.JS

EXPRESS

PROCESS AUTOMATION

Database

PERFORMANCE TUNING

Restructured PostgreSQL transactional tables with improved schemas and indexes, eliminating N+1 queries and reducing query times by an order of magnitude. MongoDB for request logs and unstructured data.

POSTGRESQL

MONGODB

QUERY OPTIMIZATION

CORE SKILLS

AI ENGINEERING

Claude Code, Sub-agent orchestration, Agentic workflows, Spec-driven development, LLM-based code review and testing, Prompt engineering, Project-level skills/hooks

BACKEND

TypeScript, NestJS, Node.js, Prisma, PostgreSQL, REST APIs, ts-rest, Zod, GraphQL, Microservices

MOBILE & WEB

React, Next.js, React Native, TanStack Query

RELIABILITY

BullMQ, Outbox pattern, Idempotency, Redis, Audit logging, Datadog

CLOUD & INFRA

AWS, GCP, Docker, CI/CD, Nx Monorepo, Git

COLLABORATION

ADRs, Cross-functional discovery, Stakeholder workshops, Mentorship

LANGUAGES

PORTUGUESE

Native

ENGLISH

C1/C2

EDUCATION

Anhanguera

BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY · JAN 2016 — JUL 2018